

WRRRA-M Foundational Architecture and First Structural Calculation

Seven Predimensional Degrees, the Dimension Filter, and Fifteen Chiral Channels

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Status	Separate from WRRRA Core 1.0

Research decision

WRRRA-M is opened as a separate research branch. It does not alter WRRRA Core 1.0. Its first task is to test whether the seven internal dimensions used in four dimensional M theory compactification can be reinterpreted as seven predimensional degrees that are rendered through the WRRRA dimension filter.

Abstract

This research note defines WRRRA-M as a testable branch of the Wonsik Reality Renderer Architecture. The branch treats the seven internal dimensions of eleven dimensional M theory as a mathematical image of seven predimensional degrees rather than as seven small spatial directions. A dimension filter renders those degrees into a three dimensional spatial phenotype, a residue ordered time direction, and particle channels. Gravity is placed at the interface because it reads the total upstream kinetic and stress structure, while gauge interactions act on labels that already passed the filter. The note also records the first nontrivial structural calculation. Two copies of the seven dimensional fundamental representation of G2 together with one invariant channel give fifteen channels. Under the color subgroup SU(3), their branching multiplicities exactly match the color content of one Standard Model generation. This is an exact color sector match, not yet a derivation of weak isospin, hypercharge, chirality, masses, or three generations.

Main result

The first WRRRA-M calculation produces an exact representation count in the color sector. The candidate channel space is one invariant channel plus two seven dimensional sectors. Each seven branches under SU(3) into a triplet, an antitriplet, and a singlet. The total therefore contains two triplets, two antitriplets, and three singlets. One Standard Model generation without a right handed neutrino has precisely the same color representation multiplicities.

$$C_{15} = \mathbf{1} \oplus \mathbf{7}_A \oplus \mathbf{7}_B$$

$$\mathbf{7} \downarrow_{SU(3)} = \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}$$

$$C_{15} \downarrow_{SU(3)} = 2 \cdot \mathbf{3} \oplus 2 \cdot \bar{\mathbf{3}} \oplus 3 \cdot \mathbf{1}$$

This result raises the WRRRA-M branch above a numerical coincidence between seven and fifteen. It supplies a concrete group theoretic bridge. The bridge remains incomplete until the filter also derives the SU(2) pairing and the U(1) hypercharges.

Scope and fixed evaluation rule

WRRRA-M follows the fixed WRRRA evaluation sequence for every claim. Known measurements and established mathematical results may calibrate the model. Once the model is fixed, outputs for unmeasured quantities count as WRRRA-M predictions. Independent prediction is not imposed as a prerequisite for theoretical value.

Stage	Question
Verification input	What measured value or established mathematical result enters the calculation
WRRRA specific transformation	What operation is performed by the WRRRA-M architecture
Output	What value structure or relation follows
Falsification condition	What observation or failed derivation would reject the claim

External physical baseline

The following statements belong to the external baseline and are not claimed as WRRRA discoveries. Superstring theories are normally formulated in ten spacetime dimensions. Witten argued that eleven dimensional supergravity appears as the low energy limit of strongly coupled type IIA string theory, providing the central route to M theory [1]. Four dimensional compactifications of M theory use seven dimensional internal spaces, often with G2 holonomy [2]. Gravity is already present in the eleven dimensional theory and is not a twelfth dimension. Singular G2 compactifications can support non Abelian gauge groups and chiral fermions in four dimensions [3][4].

WRRRA-M changes the interpretation, not these mathematical inputs. It asks whether seven geometric dimensions in the external formalism are the calculational image of seven predimensional degrees in WRRRA. A successful correspondence must reproduce more than the dimension count. It must recover the four dimensional metric, the fifteen chiral channels, their gauge assignments, and the universal coupling of gravity.

External formalism	WRRRA-M interpretation	Current status
Eleven dimensional spacetime	Four phenotype coordinates plus seven predimensional degrees	Correspondence hypothesis
Seven dimensional compact space	Calculational representation of P7	Defined but not derived
Compactification	Dimension filter projection	Architecture definition

External formalism	WRRRA-M interpretation	Current status
Eleven dimensional metric	Universal upstream kinetic structure inherited by the phenotype	Candidate bridge
Four dimensional fields	Phenotype channels selected by the filter	Open derivation

WRRRA-M ontology

The predimensional layer has no spatial coordinates and no external clock. The symbol P_7 therefore does not denote a seven dimensional place. It denotes seven independent degrees needed by the renderer before space and time are expressed. This choice preserves the WRRRA premise that the phenotype does not reveal the complete universe.

$$\mathcal{U}_M = \{S, L, P_7, R, DF_M, \Phi_{3+1}\}$$

Symbol	Meaning	Constraint
S	Source	Finite actualizable information
L	Law	Invariant transition rule
P_7	Seven predimensional degrees	No spatial interpretation is assumed
R	Residue	Present record of earlier transitions
DF_M	Meta dimensional filter	Produces finite phenotype outputs
Φ_{3+1}	Rendered spacetime and fields	Incomplete phenotype

Every phenotype point is local in the output and directly connected to the same Source and Law in the upstream architecture. This does not place a complete copy of the visible universe inside each point. It means that no phenotype location has privileged access to the substrate. Every location is a dimension filter interface.

State transition and emergent time

A state transition is indexed by n only to record order. The index is not an absolute background time. The transition rule acts on the predimensional state and its residue. A quantum or otherwise irreducible selection can enter through ξ .

$$p_{n+1} = T_L(p_n, r_n, \xi_n), \quad r_{n+1} = U_L(r_n, p_n, p_{n+1}, \xi_n)$$

The current phenotype is a filtered function of the present predimensional state and the residue available at the same transition.

$$\Phi_n(x) = DF_{M,x}(p_n, r_n)$$

Physical time appears when the renderer orders distinguishable residues. Without an ordered residue relation, an instant and an unlimited duration have no experiential distinction. The first filter target is therefore three spatial relations, one residue ordering, and fifteen chiral channels.

$$DF_M : (P_7, R) \longrightarrow (\Sigma_3, <_R, C_{15})$$

Layer architecture

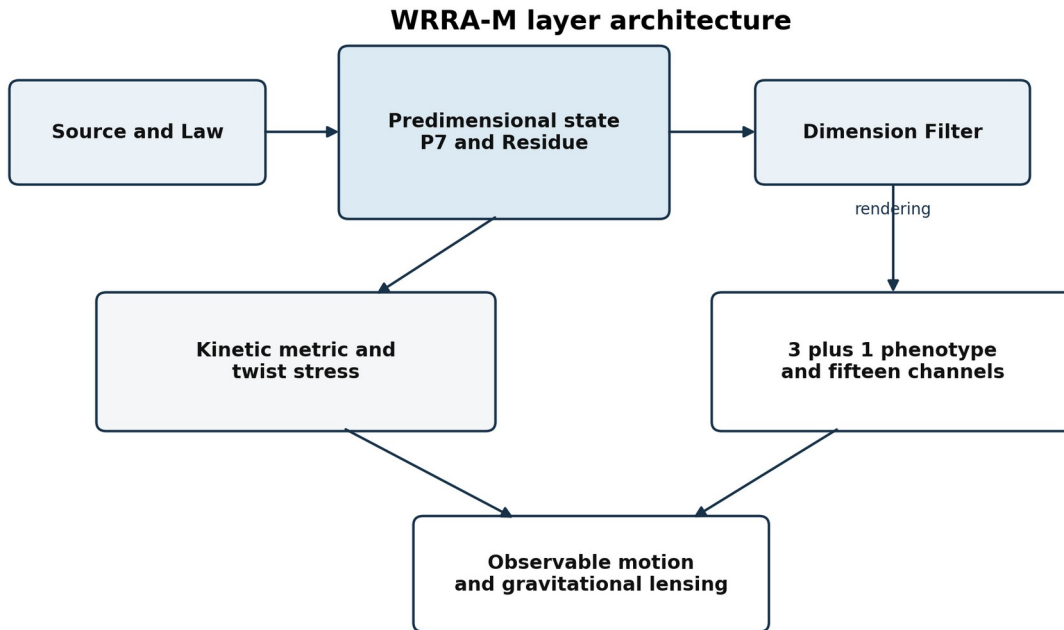


Figure 1 WRRM keeps the predimensional state separate from the phenotype and assigns gravity to the universal interface

Gravity as the universal upstream port

WRRM does not add gravity as a twelfth coordinate. Gravity is the phenotype inheritance of the total upstream kinetic and stress structure. Electromagnetic, weak, and strong interactions require labels such as charge, chirality, and color that already exist in the phenotype channel space. Gravity reads the total load before those labels are selected.

A candidate metric ansatz separates a positive spatial relation h from a residue ordering direction u . The Jacobian J connects local phenotype coordinates to the predimensional degrees after a local chart has been rendered. K is the upstream kinetic metric.

$$g_{\mu\nu}^{\text{eff}} = h_{\mu\nu} - \lambda u_\mu u_\nu, \quad h_{\mu\nu} = J_\mu^a K_{ab} J_\nu^b$$

This ansatz is a construction target, not a completed derivation. It must produce one timelike and three spacelike directions, remain stable under allowed transitions, and recover the tested weak field and relativistic limits.

Dark matter as twist stress

The same interface allows an upstream state to have zero matter phenotype and a nonzero gravitational output. WRRM names the effective source twist stress. The familiar four dimensional field equation can be used as an interface equation while the microscopic origin of the second term is calculated from P7 and the filter.

$$G_{\mu\nu}[g^{\text{eff}}] = 8\pi G_N (T_{\mu\nu}^{\text{vis}} + T_{\mu\nu}^{\text{twist}})$$

$$DF_{\text{matter}}(p_{\text{dark}}) = 0, \quad T_{\mu\nu}^{\text{twist}}(p_{\text{dark}}) \neq 0$$

The claim is stronger than saying that unknown matter gravitates. It says that the extra gravitational source is the failure of the phenotype to display the complete upstream geometry. The model must use one transferable stress law across galaxy rotation, lensing, cluster dynamics, and cosmological structure. Object by object tuning would not establish the claim.

First structural calculation from seven to fifteen

The central calculation begins with the real seven-dimensional fundamental representation of compact G2 [8]. Its complexification restricts to the SU(3) subgroup as a color triplet, an antitriplet, and a singlet [5]. WRRRA-M introduces two oriented copies of this complexified sector plus one invariant channel. The two copies are provisionally interpreted as a state sector and a residue-dual sector; their physical distinction remains to be derived from a transition law.

Representation-to-channel assumption. To interpret the fifteen complex color components as particle channels, WRRRA-M separately assigns independent complex left-handed Weyl fields to them. It does not impose the original real-representation conjugation as a reality constraint identifying the triplet and antitriplet fields. Complexification and this chirality assignment are physical modeling assumptions, not consequences of the color branching. The exact color multiplicities therefore do not derive physical chirality.

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The Standard Model color content of one generation has the same decomposition. The left handed quark doublet supplies two color triplets. The left handed conjugates of the up and down singlets supply two antitriplets. The lepton doublet and the charged lepton singlet supply three color singlets.

Standard Model field	Gauge representation	Weyl count	Candidate WRRRA-M placement
Q_L	(3, 2), Y = 1/6	6	Two triplets from 7 A and 7 B paired by the chiral filter
u ^c	(anti-3, 1), Y = -2/3	3	Antitriplet from one seven sector
d ^c	(anti-3, 1), Y = 1/3	3	Antitriplet from the other seven sector
L_L	(1, 2), Y = -1/2	2	Two singlets from 7 A and 7 B paired by the chiral filter
e ^c	(1, 1), Y = 1	1	Invariant channel

Calculated result The color representation multiplicities match exactly. Open result The present calculation does not yet derive why the triplets and two singlets pair as weak doublets while the two antitriplets remain weak singlets. It also does not derive hypercharge. Those are the next filter equations.

Null state extension

WRRRA already contains the WJNS null state. Adding one sterile null channel to the fifteen visible chiral channels gives sixteen. This mirrors the difference between the minimal Standard Model generation and a sixteen channel generation containing a sterile right handed neutrino. The match is a candidate structural interpretation and is not yet a particle identification.

$$C_{16} = C_{15} \oplus \mathbf{1}_{\text{null}}$$

Calibrated anomaly benchmark

The known Standard Model hypercharges may be used as calibration values under the fixed WRRRA evaluation rule. When the five field groups are assigned their observed hypercharges, all local gauge and mixed gravitational anomalies cancel. The cancellation is reproduced below as a benchmark that the eventual WRRRA-M charge operator must match.

Consistency condition	Calibrated calculation	Result
$SU(3)^3$	$2 - 1 - 1$	0
$SU(3)^2 U(1)$	$2 \times (1/2)(1/6) + (1/2)(-2/3) + (1/2)(1/3)$	0
$SU(2)^2 U(1)$	$3 \times (1/2)(1/6) + (1/2)(-1/2)$	0
$U(1)^3$	$6(1/6)^3 + 3(-2/3)^3 + 3(1/3)^3 + 2(-1/2)^3 + 1$	0
$\text{grav}^2 U(1)$	$6(1/6) + 3(-2/3) + 3(1/3) + 2(-1/2) + 1$	0
Global $SU(2)$	Three colored Q doublets plus one L doublet	4 doublets, even

The benchmark confirms consistency after calibration. It does not show that WRRRA-M has derived hypercharge. The required next result is a filter operator whose eigenvalues produce the calibrated charges and whose structure explains the cancellation rather than inserting each charge separately.

Claim ledger

Verification input	WRRRA specific transformation	Output	Falsification condition
M theory uses seven internal dimensions for a four dimensional compactification	Replace ontic internal space with seven predimensional degrees represented geometrically for calculation	P7 correspondence hypothesis	A four dimensional reduction requires structures that cannot be represented by seven predimensional degrees
G2 fundamental branching gives 7 equals 3 plus 3 bar plus 1	Use two oriented seven sectors and one invariant channel	Two triplets two antitriplets and three singlets	The filter cannot preserve the branching or cannot assign the channels to one generation
One Standard Model generation has fifteen Weyl channels	Match only the SU 3 color multiplicities	Exact color sector multiplicity match	Any channel count or color representation fails after a complete field assignment
Observed hypercharges	Calibrate the candidate charge operator and run anomaly sums	All benchmark anomalies vanish	No single filter operator can generate the charges without channel by channel insertion
Additional gravity is inferred without light	Set matter phenotype to zero while retaining twist stress	Effective dark gravitational source	A non gravitational particle signal explains all effects or one twist law cannot fit rotation lensing clusters and cosmology
Residues are ordered in the present	Use residue order as the timelike filter direction	Emergent time candidate	Lorentzian causality or observed clock relations cannot be recovered

Claim status

Code	Meaning	WRRR-M item
E	Established external input	Eleven dimensional low energy limit seven dimensional compactification G2 chirality at singularities
D	Architecture definition	Predimensional P7 residue dimension filter phenotype separation
C	Calculated within the branch	The SU 3 multiplicity equality between 1 plus 7 plus 7 and one Standard Model generation
H	Testable hypothesis	Gravity as the sole universal upstream port and dark matter as twist stress
O	Open derivation	SU 2 pairing hypercharge three generations masses and numerical dark stress law

Next calculations

1. Define the two oriented seven sectors from the transition and residue equations instead of postulating two copies.
2. Construct a dimension filter that pairs the two triplets into Q L and the two appropriate singlets into L L while leaving the antitriplets as weak singlets.
3. Derive one hypercharge operator on the full fifteen channel space. Calibrate only its minimum free parameters and reproduce all anomaly cancellations.
4. Test whether the WJNS null state naturally has the sterile quantum numbers of a right handed neutrino and whether its activation generates a neutrino mass scale.
5. Specify the kinetic metric K and derive a Lorentzian three plus one metric with the residue order as the timelike direction.
6. Derive a transferable twist stress profile and test it against galaxy rotation gravitational lensing cluster dynamics and cosmological structure with one parameter policy.
7. Map the established WRRR fifteen channel implementation to the new one plus seven plus seven basis and identify whether the two descriptions are equivalent or one is more restrictive.

Minimum success conditions

- The filter produces rank four local spacetime with one timelike direction.
- The channel projector has rank fifteen and recovers the full Standard Model gauge assignments for one generation.
- All gauge and mixed anomalies cancel from the derived operator.
- Gravity couples to the complete upstream load while the other interactions remain current selective.
- One twist stress law covers the main dark matter observables without object specific tuning.
- The null state extension either yields a consistent sterile channel or is rejected.

Conclusion

WRRRA-M begins with one exact structural result and several explicit open problems. The exact result is the color sector identity between one invariant channel plus two G2 seven sectors and the fifteen Weyl channels of one Standard Model generation. The open problems are decisive. Weak pairing, hypercharge, chirality, generations, mass, and the numerical twist stress law must be derived. The branch will remain separate from WRRRA Core 1.0 until those tests are passed.

The working interpretation is concise. The seven internal dimensions of M theory may be the geometric calculation of seven predimensional degrees. The dimension filter renders their finite phenotype. Gravity is the inherited metric of the full upstream state. Dark matter is the gravitational remainder of what the phenotype does not display.

Appendix A First calculation in compact form

$$C_{15} = \mathbf{1} \oplus \mathbf{7}_A \oplus \mathbf{7}_B$$

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Color content	WRRM-M count	Standard Model one generation
Triplet	2	Q L contains two
Antitriplet	2	u c and d c
Singlet	3	Two in L L and one e c
Total Weyl channels	15	15

Interpretive boundary The equality is exact under SU 3 color restriction. It is not yet an equality of full SU 3 times SU 2 times U 1 representations.

Reproduction: calculations/verify_wrra_m_0_1.py checks the SU(3) torus characters, the 15-component color count, and the calibrated anomaly benchmark with exact arithmetic. It treats the G2 branching as an established input and does not test physical chirality.

Appendix B References

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